Algebra & Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Expanding and simplifying expressions

Expand and simplify $3(2x - 1) - 2(x - 4)$.

$6x - 3 - 2x + 8 = 4x + 5$.

2 [Foundational] • Target: Factoring quadratic expressions

Factor $x^2 - 49$ using the difference of squares.

$(x - 7)(x + 7)$.

3 [Foundational] • Target: The quadratic formula and the discriminant

Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve $x^2 + 3x + 1 = 0$, leaving the answer in exact form.

$a = 1, b = 3, c = 1$. $x = \frac{-3 \pm \sqrt{9 - 4}}{2} = \frac{-3 \pm \sqrt{5}}{2}$.

4 [Foundational] • Target: The quadratic formula and the discriminant

Find the discriminant of $2x^2 - 4x + 5 = 0$ and state how many real roots it has.

$\Delta = (-4)^2 - 4(2)(5) = 16 - 40 = -24$. Since $\Delta < 0$, there are no real roots.

5 [Foundational] • Target: The quadratic formula and the discriminant

Use the quadratic formula to solve $3x^2 - 2x - 1 = 0$.

$\Delta = (-2)^2 - 4(3)(-1) = 4 + 12 = 16$. $x = \frac{2 \pm 4}{6}$, so $x = 1$ or $x = -\frac{1}{3}$.

6 [Foundational] • Target: Linear inequalities

Solve $2x - 5 < 9$ and graph the solution set on a number line.

$2x < 14 \Rightarrow x < 7$.

7 [Foundational] • Target: Linear inequalities

Solve the compound inequality $5 < 2x + 1 \leq 11$.

Subtract 1: $4 < 2x \leq 10$. Divide by 2: $2 < x \leq 5$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Systems of linear equations

Solve the system $x + y = 10$, $x - y = 2$.

Adding the equations: $2x = 12 \Rightarrow x = 6$. Then $y = 10 - 6 = 4$.

9 [Intermediate] • Target: Systems of linear equations

This question has two parts. The sum of two numbers is 15 and their difference is 3.

a Write a system of two equations for the numbers x and y. $x + y = 15$ and $x - y = 3$.

b Solve the system to find both numbers. Adding the equations: $2x = 18 \Rightarrow x = 9$. Then $y = 15 - 9 = 6$.

(a) $x + y = 15$ and $x - y = 3$.; (b) Adding the equations: $2x = 18 \Rightarrow x = 9$. Then $y = 15 - 9 = 6$.

10 [Intermediate] • Target: Solving for a variable in a formula

The area of a triangle is $A = \frac{1}{2}bh$. Solve for h.

$h = \frac{2A}{b}$.

11 [Intermediate] • Target: Solving for a variable in a formula

The circumference of a circle is $C = 2\pi r$. Solve for r.

$r = \frac{C}{2\pi}$.

12 [Intermediate] • Target: Solving for a variable in a formula

The formula $v = v_0 + at$ gives velocity under constant acceleration. Solve for t.

$t = \frac{v - v_0}{a}$.

13 [Intermediate] • Target: Word problems: setting up and solving equations

This question has two parts. A rectangular garden has a length that is 4 meters more than its width w. Its area is 96 square meters.

a Write an equation relating w to the area. $w(w + 4) = 96$, i.e. $w^2 + 4w - 96 = 0$.

b Solve for w, and state the garden's length. Factoring: $(w - 8)(w + 12) = 0$. Since width is positive, $w = 8$ meters, so the length is $w + 4 = 12$ meters.

(a) $w(w + 4) = 96$, i.e. $w^2 + 4w - 96 = 0$.; (b) Factoring: $(w - 8)(w + 12) = 0$. Since width is positive, $w = 8$ meters, so the length is $w + 4 = 12$ meters.

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14 [Intermediate] • Target: Word problems: setting up and solving equations

This question has two parts. A mobile-data plan charges a fixed monthly fee f plus a rate r per gigabyte used. One month, 5 GB of data cost a total of 65 dollars; another month, 9 GB cost a total of 93 dollars.

a Write a system of two equations for f and r.

$f + 5r = 65$ and $f + 9r = 93$.

b Solve the system to find the fixed fee and the rate per gigabyte. Subtracting the first equation from the second: $4r = 28 \Rightarrow r = 7$. Then $f = 65 - 5(7) = 30$. Fixed fee: 30 dollars; rate: 7 dollars per GB.

(a) $f + 5r = 65$ and $f + 9r = 93$.; (b) Subtracting the first equation from the second: $4r = 28 \Rightarrow r = 7$. Then $f = 65 - 5(7) = 30$. Fixed fee: 30 dollars; rate: 7 dollars per GB.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: MCQ — Factoring quadratic expressions • Cross-Lesson: 9.01 Statistics & Probability

Find the value of m such that the quadratic equation $x^2 - 6x + m = 0$ has exactly one repeated real root.

Step 1: Set discriminant to zero: Exactly one repeated real root requires $\Delta = b^2 - 4ac = 0$. Step 2: Substitute coefficients and solve: Here $a = 1$, $b = -6$, $c = m$:

$(-6)^2 - 4(1)(m) = 0 \Rightarrow 36 - 4m = 0 \Rightarrow 4m = 36 \Rightarrow \mathbf{m = 9}$.

16 [Challenging] • Target: MCQ — Solving quadratic equations by factoring • Cross-Lesson: 2.01 Functions & Transformations

If $(x - 2)$ is a factor of the polynomial $P(x) = 2x^3 - 3x^2 + kx - 10$, determine the value of k.

Step 1: Apply the Factor Theorem: Since $(x - 2)$ is a factor, the remainder upon division by $(x - 2)$ is zero, meaning $P(2) = 0$. Step 2: Evaluate $P(2)$ and solve for k: $P(2) = 2(2)^3 - 3(2)^2 + k(2) - 10 = 0$

$2(8) - 3(4) + 2k - 10 = 0 \Rightarrow 16 - 12 + 2k - 10 = 0 \Rightarrow 2k - 6 = 0 \Rightarrow \mathbf{k = 3}$.

17 [Challenging] • Target: MCQ — The quadratic formula and the discriminant • Cross-Lesson: 3.01 Sequences & Series

For what value of k is the matrix $A = (k, 4; 3, 6)$ singular (has no inverse)?

Step 1: State singularity criterion: A square matrix is singular if its determinant equals zero: $\det(A) = 0$. Step 2: Compute determinant and solve: $\det(A) = (k)(6) - (4)(3) = 6k - 12 = 0 \Rightarrow 6k = 12 \Rightarrow \mathbf{k = 2}$.

18 [Challenging] • Target: MCQ — Absolute value equations • Cross-Lesson: 4.01 Trigonometry

Find the solution set of the inequality $|3x - 1| > 8$.

Step 1: Split into disjoint cases: The absolute value inequality $|u| > k$ splits into $u > k$ or x.Case 2: \(3x - 1

Step 2: Express in interval notation: $\mathbf{(-\infty, -\frac{7}{3}) \cup (3, \infty)}$.

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19 [Challenging] • Target: MCQ — Linear inequalities • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

If $f(x) = x^2 + 1$ and $g(x) = \sqrt{x}$, determine the formula for $(f \circ g)(x)$ and state its domain.

Step 1: Compute composite formula: $(f \circ g)(x) = f(g(x)) = f(\sqrt{x}) = (\sqrt{x})^2 + 1 = x + 1$. Step 2: Determine domain: Although $x + 1$ is defined for all real numbers, the inner function $g(x) = \sqrt{x}$ requires $x \geq 0$. Therefore, the domain of the composite function is: **$[0, \infty)$**.

20 [Challenging] • Target: MCQ — Systems of linear equations • Cross-Lesson: 6.01 Limits & Derivatives

Determine the horizontal and vertical asymptotes of the rational function $g(x) = \frac{4x-1}{2x+6}$.

Step 1: Vertical asymptote: Set the denominator to zero: $2x + 6 = 0 \Rightarrow 2x = -6 \Rightarrow$ **$x = -3$**. Step 2: Horizontal asymptote: Since the degree of numerator (1) equals the degree of denominator (1), take the ratio of leading coefficients: $y = \frac{4}{2} =$ **2**.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — The quadratic formula and the discriminant

Use the quadratic formula to solve $x^2 + 5x + 2 = 0$, leaving the answer in surd form.

a. $x = \frac{5 \pm \sqrt{17}}{2}$

b. $x = \frac{-5 \pm \sqrt{17}}{4}$

c. $x = \frac{-5 \pm \sqrt{33}}{2}$

d. $x = \frac{-5 \pm \sqrt{17}}{2}$ ✓

Discriminant $= 25 - 8 = 17$; $x = \frac{-5 \pm \sqrt{17}}{2}$.

22 [Foundational] • Target: MCQ — Absolute value equations

The graph shows $y = |x - 3|$. At what x-value does the graph reach its minimum?

a. $x = 1$

b. $x = 3$ ✓

c. $x = -3$

d. $x = 0$

The vertex of the V-shape (the minimum) is at $x = 3$.

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23 [Foundational] • Target: MCQ — Linear inequalities

Solve $3x - 7 < 14$, and identify the graph of its solution set.

a. $x < 7$ ✓

b. $x \geq 7$

c. $x \leq 7$

d. $x > 7$

$3x < 21 \Rightarrow x < 7$.

24 [Foundational] • Target: MCQ — Linear inequalities

Solve $2x + 5 > -3$ and $x - 1 < 6$ together (both must hold). Find the combined solution set.

a. $x > -4$

b. $-4 < x \leq 7$

c. $x < 7$

d. $-4 < x < 7$ ✓

$2x + 5 > -3 \Rightarrow x > -4$; $x - 1 < 6 \Rightarrow x < 7$. Together: $-4 < x < 7$.

25 [Foundational] • Target: MCQ — Systems of linear equations

The sum of two numbers is 22 and their difference is 6. Find the two numbers.

a. 11 and 11

b. 14 and 6

c. 16 and 6

d. 14 **and** 8 ✓

$x + y = 22, x - y = 6 \Rightarrow x = 14, y = 8$.

26 [Foundational] • Target: MCQ — Solving for a variable in a formula

The perimeter of a rectangle is $P = 2l + 2w$. Solve for l.

a. $l = \frac{P}{2} - w$

b. $l = \frac{P - 2w}{2}$ ✓

c. $l = \frac{P + 2w}{2}$

d. $l = P - 2w$

$P - 2w = 2l \Rightarrow l = \frac{P - 2w}{2}$ (equivalently $\frac{P}{2} - w$).

27 [Foundational] • Target: MCQ — Solving for a variable in a formula

The formula for simple interest is $I = Prt$. Solve for r.

a. $r = \frac{I}{P}$

b. $r = \frac{I}{Pt}$ ✓

c. $r = \frac{Pt}{I}$

d. $r = I - Pt$

Dividing both sides by Pt: $r = \frac{I}{Pt}$.

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28 [Foundational] • Target: MCQ — Solving for a variable in a formula

The formula $F = \frac{9}{5}C + 32$ converts Celsius to Fahrenheit. Solve for C.

a. $C = \frac{5(F-32)}{9}$ ✓

b. $C = \frac{5F-32}{9}$

c. $C = \frac{5}{9}F + 32$

d. $C = \frac{9(F-32)}{5}$

$F - 32 = \frac{9}{5}C \Rightarrow C = \frac{5(F-32)}{9}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Solving for a variable in a formula

The volume of a cylinder is $V = \pi r^2 h$. Solve for h.

a. $h = \frac{V}{\pi r^2}$ ✓

b. $h = V - \pi r^2$

c. $h = \frac{V}{\pi r}$

d. $h = \frac{\pi r^2}{V}$

Dividing both sides by πr^2: $h = \frac{V}{\pi r^2}$.

30 [Intermediate] • Target: MCQ — Linear-quadratic systems

The graph shows the system $y = x^2 - 4$, $y = 3x$. What are the points of intersection?

a. $(4, 12)$ and $(1, 3)$

b. $(-4, -12)$ and $(1, 3)$

c. $(-1, -3)$ and $(1, 3)$

d. $(4, 12)$ **and** $(-1, -3)$ ✓

The curve and line cross at $(-1, -3)$ and $(4, 12)$.

31 [Intermediate] • Target: MCQ — Word problems: setting up and solving equations

A rectangular garden has a length that is 5 meters more than its width, and its area is 84 square meters. Find the width.

a. 12

b. 7 ✓

c. 9

d. 14

$w(w + 5) = 84 \Rightarrow (w + 12)(w - 7) = 0$; the positive root is $w = 7$.

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32 [Intermediate] • Target: MCQ — Word problems: setting up and solving equations

A mobile-data plan charges a fixed monthly fee of 40 riyals plus 2 riyals per GB used. If a user's bill was 68 riyals, how many GB did they use?

a. 34

b. 14 ✓

c. 54

d. 28

$40 + 2g = 68 \Rightarrow 2g = 28 \Rightarrow g = 14$.

33 [Intermediate] • Target: MCQ — Solving linear equations

[illegible] $ax^2 + bx + c = 0$ [illegible] $\Delta = b^2 - 4ac$ [illegible] $2x^2 - 8x + k = 0$ [illegible] k [illegible] For a quadratic equation $ax^2 + bx + c = 0$, if the discriminant $\Delta = b^2 - 4ac$ is equal to zero, it means the equation has one repeated real root. Given the equation $2x^2 - 8x + k = 0$, what is the value of k that results in this equation having one repeated real root?

a. 4

b. 8

c. **16** ✓

d. 32

[illegible] $2x^2 - 8x + k = 0$ [illegible] $\Delta = b^2 - 4ac = 0$. [illegible] $a = 2$ [illegible] $b = -8$ [illegible] $c = k$. [illegible]: $(-8)^2 - 4(2)(k) = 0$ $64 - 8k = 0$ $64 = 8k$ $k = \frac{64}{8}$ $k = 8$ For the quadratic equation $2x^2 - 8x + k = 0$ to have one repeated real root, the discriminant $\Delta = b^2 - 4ac$ must be equal to 0. Here, $a = 2$, $b = -8$, and $c = k$. Substitute the values into the discriminant formula: $(-8)^2 - 4(2)(k) = 0$ $64 - 8k = 0$.

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34 [Intermediate] • Target: MCQ — Expanding and simplifying expressions

■■■■■■ ■■■■■■■■■■ $f(x) = ax^2 + bx + c$■ ■■■■ ■■■■■ ■■ $f(0) = 5$ ■■■■ $f(1) = 8$ ■■■■ $f(2) = 13$■ ■■■■■■ ■■■■■ a. For the quadratic function $f(x) = ax^2 + bx + c$, if $f(0) = 5$, $f(1) = 8$, and $f(2) = 13$, find the value of a.

a. **1** ✓

b. 2

c. 3

d. 4

$f(0) = a(0)^2 + b(0) + c = c = 5$. ■■■■■ $f(x) = ax^2 + bx + 5$. $f(1) = a(1)^2 + b(1) + 5 = a + b + 5 = 8$ ■■■■ ■■■■■ ■■ $a + b = 3$. (1) $f(2) = a(2)^2 + b(2) + 5 = 4a + 2b + 5 = 13$ ■■■■ ■■■■■ ■■ $4a + 2b = 8$■ ■■■■■■■■■ ■■■■ 2■ ■■■■■ ■■■■ $2a + b = 4$. (2) ■■■■■ ■■■■■■■■■ (1) ■■ ■■■■■■■■■ (2): $(2a + b) - (a + b) = 4 - 3$ $a = 1$ $f(0) = a(0)^2 + b(0) + c = c = 5$. So, $f(x) = ax^2 + bx + 5$. $f(1) = a(1)^2 + b(1) + 5 = a + b + 5 = 8$ This implies $a + b = 3$. (1) $f(2) = a(2)^2 + b(2) + 5 = 4a + 2b + 5 = 13$ This implies.

35 [Intermediate] • Target: MCQ — Factoring quadratic expressions

■■ ■■ ■■■■■■ ■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ $x^2 - 5x + 6 = 0$■ What is the sum of the roots of the quadratic equation $x^2 - 5x + 6 = 0$?

a. 6

b. -5

c. **5** ✓

d. -6

■■■■■■ ■■■■■■ ■■■■■■■ ■■■■ ■■■■■■ ■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ $ax^2 + bx + c = 0$ ■■ $-\frac{b}{a}$. ■■ ■■■■ ■■■■■■■■■■ $a = 1$ ■ $b = -5$. ■■■■■■ ■■■■■■ ■■■■■■■ ■■ $-\frac{-5}{1} = 5$. According to Vieta's formulas, the sum of the roots of a quadratic equation $ax^2 + bx + c = 0$ is $-\frac{b}{a}$. In this equation, $a = 1$ and $b = -5$. Therefore, the sum of the roots is $-\frac{-5}{1} = 5$.

36 [Intermediate] • Target: MCQ — Solving quadratic equations by factoring

■■ ■■■■■ ■■■■■■ ■■■■■■■■■ $x^2 + 5x + 3 = 0$■ What is the discriminant of $x^2 + 5x + 3 = 0$?

a. **13** ✓

b. 19

c. 18

d. 11

■■■■■■■ ■■■■■■■■ ■■■■■■■■■ ■■ Algebra & Equations: $\Delta = b^2 - 4ac = (5)^2 - 4(1)(3) = 13$ ■■■■■■■■■ ■■■■ ■■■■■■■■ ■■■■■■■■ ■■ 13. Applying Algebra & Equations principles: $\Delta = b^2 - 4ac = (5)^2 - 4(1)(3) = 13$ Therefore, the correct choice is 13.

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37 [Intermediate] • Target: MCQ — The quadratic formula and the discriminant

[redacted] If one root of a real quadratic equation is $2 + 3i$, what is the other root?

a. $2 - 3i$ ✓

b. $-2 + 3i$

c. $-2 - 3i$

d. $3 + 2i$

[redacted].

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Absolute value equations • Cross-Lesson: 4.01 Trigonometry

[redacted] The function $f(x) = x^2 - 6x + 9$ represents the trajectory of a projectile. What is the minimum height the object reaches?

a. **0** ✓

b. 3

c. -3

d. 9

[redacted].

39 [Challenging] • Target: MCQ — Linear inequalities • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

[redacted] $y = mx + c$ [redacted] y [redacted] $(0, 5)$ [redacted] $(2, 11)$. [redacted] m[redacted] [redacted] [redacted] In the figure below, the straight line $y = mx + c$ represents a line that intersects the y-axis at the point $(0, 5)$ and passes through the point $(2, 11)$. What is the value of m, the slope of the line?

a. -3

b. **3** ✓

c. -2

d. 2

[redacted] $y = mx + c$[redacted] [redacted] m [redacted] c [redacted] y. [redacted] y [redacted] $(0, 5)$[redacted] [redacted] $c = 5$. [redacted] $y = mx + 5$. [redacted] $(2, 11)$. [redacted] m: $11 = m(2) + 5\ 11 - 5 = 2m\ 6 = 2m\ m = \frac{6}{2}\ m = 3$ We know that the equation of a straight line is $y = mx + c$, where m is the slope and c is the y-intercept. From the given information, the line int.

40 [Challenging] • Target: MCQ — Systems of linear equations • Cross-Lesson: 6.01 Limits & Derivatives

[redacted] $y = mx + c$ [redacted] x [redacted] y. [redacted] $\frac{1}{2}$ [redacted] $(4, 5)$[redacted] [redacted] c. In the diagram below, the straight line $y = mx + c$ intersects the x and y axes. If its slope is $\frac{1}{2}$ and it passes through the point $(4, 5)$, find the value of the constant term c.

a. 3

b. -3 ✓

c. 1

d. -1

[redacted] $y = mx + c$[redacted] [redacted] m [redacted] c [redacted] [redacted] y). [redacted] $m = \frac{1}{2}$ [redacted] $(4, 5)$. [redacted] c: $5 = \left(\frac{1}{2}\right)(4) + c$ $5 = 2 + c$ [redacted] 2 [redacted]: $c = 5 - 2\ c = 3$ The general equation of a straight line is $y = mx + c$, where m is the slope and c is the constant term (y-intercept). We know that the slope $m = \frac{1}{2}$ and the line passes through the poin.

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41 [Challenging] • Target: MCQ — Rational equations • Cross-Lesson: 7.01 Integration & Applications

[illegible] [illegible] [illegible] [illegible] $f(x) = x^2 - 4x + 3$ [illegible] $g(x) = x - 1$. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] (x, y) [illegible] $y = 0$[illegible] The graph shows the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. What are the coordinates of the intersection point (x, y) where $y = 0$?

a. $(1, 0)$ ✓

b. $(3, 0)$

c. $(0, 3)$

d. $(0, 1)$

[illegible] [illegible] [illegible] [illegible] [illegible] $f(x)$ [illegible] $g(x)$: $x^2 - 4x + 3 = x - 1$. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: $x^2 - 5x + 4 = 0$. [illegible] [illegible] [illegible]: (x-1)(x-4) = 0 \>. [illegible] x=1 \) [illegible] $x = 4$. [illegible] x=1 \)[illegible] $y = g(1) = 1 - 1 = 0$. [illegible] x=4 \)[illegible] $y = g(4) = 4 - 1 = 3$. [illegible] (1, 0) \) [illegible] [illegible] [illegible] [illegible] $y = 0$. To find the intersection points of the two functions, we set f(x) \) equal to $g(x)$: $x^2 - 4x + 3 = x - 1$. Rearrange the equation to one side: $x^2 - 5x + 4 = 0$. Factor the quadratic equation: (x-1)(x-4) =.

42 [Challenging] • Target: MCQ — Solving for a variable in a formula • Cross-Lesson: 8.01 Exponentials & Logarithms

[illegible] [illegible] x [illegible] y [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: $\{x^2 + y^2 = 13\ xy = 6$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $x + y$[illegible] [illegible] [illegible] x [illegible] y [illegible] [illegible] [illegible] If x and y are real numbers satisfying the following system of equations: $\{x^2 + y^2 = 13\ xy = 6$ What is the maximum possible value of the expression $x + y$, given that x and y have the same sign?

a. 3

b. **5** ✓

c. 6

d. 7

[illegible] [illegible] $(x + y)^2 = x^2 + 2xy + y^2$. [illegible] [illegible] [illegible] [illegible] [illegible]: $(x + y)^2 = (x^2 + y^2) + 2xy\ (x + y)^2 = 13 + 2(6)\ (x + y)^2 = 13 + 12\ (x + y)^2 = 25$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: $x + y = \pm\sqrt{25}\ x + y = \pm 5$ [illegible] [illegible] x [illegible] y [illegible] [illegible] [illegible] [illegible] [illegible] $xy = 6$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] x [illegible] y [illegible] [illegible] $x + y$ [illegible] [illegible] [illegible] $x + y = 5$. [illegible] [illegible] x [illegible] y [illegible] [illegible] $x + y$ [illegible] [illegible] [illegible] $x + y = -5$. [illegible] [illegible] [illegible] [illegible] [illegible] $x + y$ [illegible] 5. We know.

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43 [Challenging] • Target: MCQ — Linear-quadratic systems • Cross-Lesson: 9.01 Statistics & Probability

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] L_1 [illegible] L_2 [illegible] [illegible] T. [illegible] [illegible] [illegible] [illegible] α [illegible] 60° [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] β[illegible] [illegible] [illegible] β [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] α[illegible] In the figure shown, you have two parallel lines L_1 and L_2 intersected by a transversal T. If the measure of angle α is 60°, what is the measure of angle β, given that β is the consecutive interior angle to α?

a. 30°

b. 60°

c. **120° ✓**

d. 90°

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 180 [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] α [illegible] β [illegible] [illegible] [illegible] [illegible] [illegible]. When a transversal intersects two parallel lines, consecutive interior angles are supplementary, meaning their sum is 180 degrees. In this case, angles α and β are consecutive interior angles. $\alpha + \beta = 180^\circ$ $\alpha + \beta = 180^\circ$ [illegible] [illegible] $\alpha = 60^\circ$ [illegible] [illegible]: Since $\alpha = 60^\circ$, th.

44 [Challenging] • Target: MCQ — Word problems: setting up and solving equations • Cross-Lesson: 2.01 Functions & Transformations

[illegible] [illegible] $5x - 7 = 3x + 9$[illegible] [illegible] [illegible] [illegible] x[illegible] [illegible] [illegible] x [illegible] [illegible] If $5x - 7 = 3x + 9$, what is the value of x, given that x is a real number?

a. 7

b. **8 ✓**

c. 6

d. 9

[illegible] [illegible] $5x - 7 = 3x + 9$[illegible] [illegible] $3x$ [illegible] [illegible] [illegible]: $5x - 3x - 7 = 3x - 3x + 9$ $2x - 7 = 9$ [illegible] [illegible] 7 [illegible] [illegible] [illegible]: $2x - 7 + 7 = 9 + 7$ $2x = 16$ [illegible] [illegible] [illegible] [illegible] [illegible] 2: $\frac{2x}{2} = \frac{16}{2}$ $x = 8$ To solve the equation $5x - 7 = 3x + 9$, we subtract $3x$ from both sides: $5x - 3x - 7 = 3x - 3x + 9$ $2x - 7 = 9$ Then we add 7 to both sides: $2x - 7 + 7 = 9 + 7$ $2x = 16$ Finally, we divide both sides by 2: $\frac{2x}{2} = \frac{16}{2}$ $x = 8$

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45 [Challenging] • Target: Solving linear equations • Cross-Lesson: 3.01 Sequences & Series

[redacted] $3x - 5 = 7$[redacted] [redacted] [redacted] x[redacted] To solve the equation $3x - 5 = 7$, what is the value of x?

a. **x = 4 ✓**

b. x = 3

c. x = 2

d. x = 5

[redacted] $3x - 5 = 7$[redacted] [redacted] [redacted] [redacted]: 1. [redacted] 5 [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] [redacted] x: $3x - 5 + 5 = 7 + 5$ $3x = 12$ 2. [redacted] [redacted] [redacted] [redacted] 3 [redacted] [redacted] x: $\frac{3x}{3} = \frac{12}{3}$ $x = 4$ To solve the equation $3x - 5 = 7$, we follow these steps: 1. Add 5 to both sides of the equation to isolate the term with x: $3x - 5 + 5 = 7 + 5$ $3x = 12$ 2. Divide both sides by 3 to find the value of x: $\frac{3x}{3} = \frac{12}{3}$ $x = 4$

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